

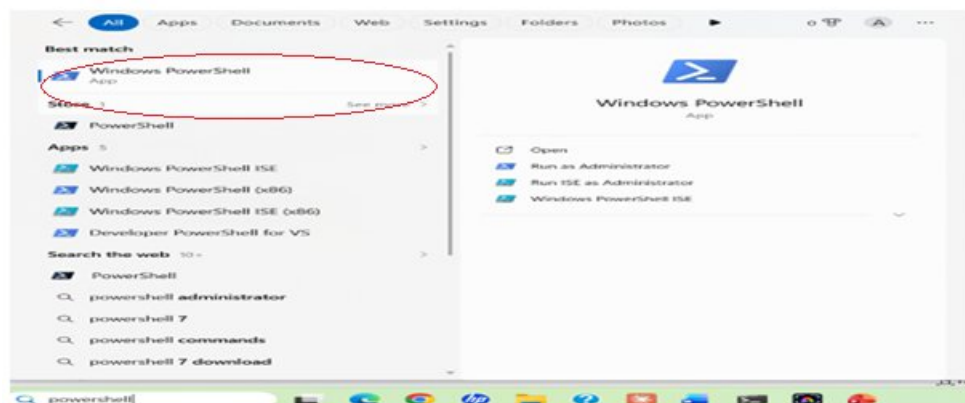
SKILL – 1

- To Install Linux VM, Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture, Build Initial Makefile

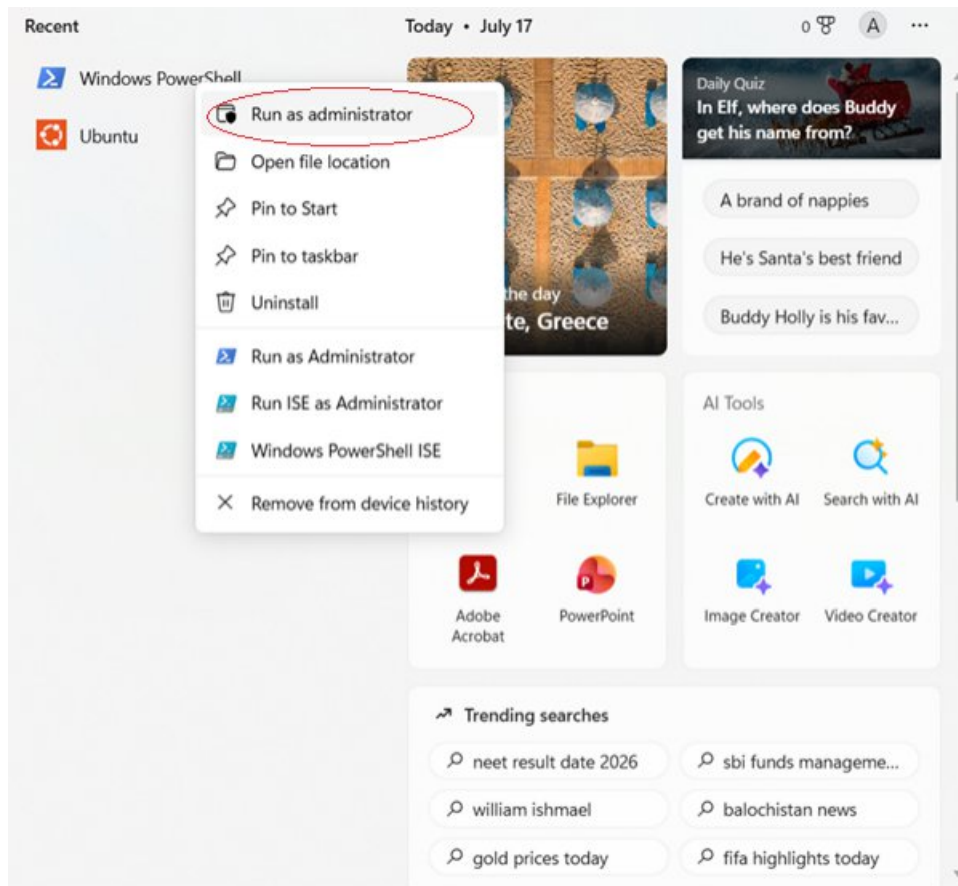
To Analyze process abstraction, execute `fork()`, understand `exec()` family, analyze parent-child relationships, inspect process tree, practice system call tracing

Step 1: Open PowerShell as Administrator

1. Press **Windows**.
2. Search for **PowerShell**.



3. Right-click **Windows PowerShell**.



```
Windows PowerShell
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PS C:\windows\system32> wsl --install
```

Step 3: Restart Your Computer

After installation, restart your PC when prompted.

Step 4: Complete Ubuntu Setup

After restarting:

1. Ubuntu opens automatically (or search for **Ubuntu** in the Start menu).
2. Wait for the installation to finish.
3. Create a Linux username.

Example:

Enter new UNIX username: Shannu

Step 5: Create a Password

You'll then be prompted to create a password.

New password: ****

Retype new password:****

Step 6: Update Ubuntu

```
PS C:\Users\Saiteja> wsl
saiteja@LAPTOP-5BAATM70:/mnt/c/Users/Saiteja$ sudo apt update
[sudo: authenticate] Password:
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [424 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [548 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [97.9 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.7 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [127 kB]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [159 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [98.0 kB]
Get:12 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [251 kB]
Get:13 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.2 kB]
Get:14 http://archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [83.6 kB]
Get:15 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [361 kB]
Get:16 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [188 kB]
Get:17 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [371 kB]
Get:18 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [70.0 kB]
Get:19 http://archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [71.7 kB]
Get:20 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Packages [3132 B]
Get:21 http://archive.ubuntu.com/ubuntu resolute-backports/universe Translation-en [7988 B]
Get:22 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Components [1056 B]
Fetched 3364 kB in 9s (378 kB/s)
25 packages can be upgraded. Run 'apt list --upgradable' to see them.
saiteja@LAPTOP-5BAATM70:/mnt/c/Users/Saiteja$
```

Step 7: Install GCC (C Compiler)

```
25 packages can be upgraded. Run 'apt list --upgradable' to see them.
saiteja@LAPTOP-5BAATM70:/mnt/c/Users/Saiteja$ sudo apt install build-essential
build-essential is already the newest version (12.12ubuntu2.26.04.2).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 25
saiteja@LAPTOP-5BAATM70:/mnt/c/Users/Saiteja$
```

Step 8: Verify the Installation

```
saiteja@LAPTOP-5BAATM70:/mnt/c/Users/Saiteja$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
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This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

saiteja@LAPTOP-5BAATM70:/mnt/c/Users/Saiteja$
```

Step 9: Create Your C Program

Example: nano process.c

Step 10: Compile the Program

```
Windows PowerShell
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PS C:\Users\Saiteja> wsl
saiteja@LAPTOP-5BAATM70:/mnt/c/Users/Saiteja$ |
```

```
saiteja@LAPTOP-5BAATM70:~/TeacherFile$ gcc hello16.c -o hello16
saiteja@LAPTOP-5BAATM70:~/TeacherFile$ ./hello16
Producer produced: 1
Consumer consumed: 1
Producer produced: 2
Consumer consumed: 2
Producer produced: 3
Consumer consumed: 3
Producer produced: 4
Consumer consumed: 4
Producer produced: 5
Consumer consumed: 5
saiteja@LAPTOP-5BAATM70:~/TeacherFile$ |
```

```
shannu@Shannu: ~/skill  GNU nano 6.2  simplecommandloop.c *
#include <stdio.h>
#include <unistd.h>
#include <string.h>

int main() {
    char command[100];

    while (1) {
        printf("ShannuVaranasi_SHELLFORGE$ ");
        fgets(command, sizeof(command), stdin);
        command[strcspn(command, "\n")] = '\0';
        if (strcmp(command, "exit") == 0)
            break;

        printf("You entered %s\n", command);
    }

    return 0;
}

File Name to Write: simplecommandloop.c
^G Help      M-D DOS Format  M-A Append     M-B Backup File
^C Cancel    M-M Mac Format  M-D Prepend    ^T Browse
repository  sum            torrent        sum            torrent        sum
```

